Laboratory Activity: Implementing OAuth 2.0 in PHP

Objective:
Students will integrate **Google OAuth 2.0** into a PHP web application running on XAMPP.

Learning Outcomes:

1. Understand OAuth 2.0 concepts: Client ID, Client Secret, Redirect URI, Authorization Code.
2. Set up a PHP server using XAMPP for OAuth testing.
3. Implement OAuth 2.0 authorization code flow in PHP.
4. Retrieve and display user profile information.

Part 1: Set up XAMPP

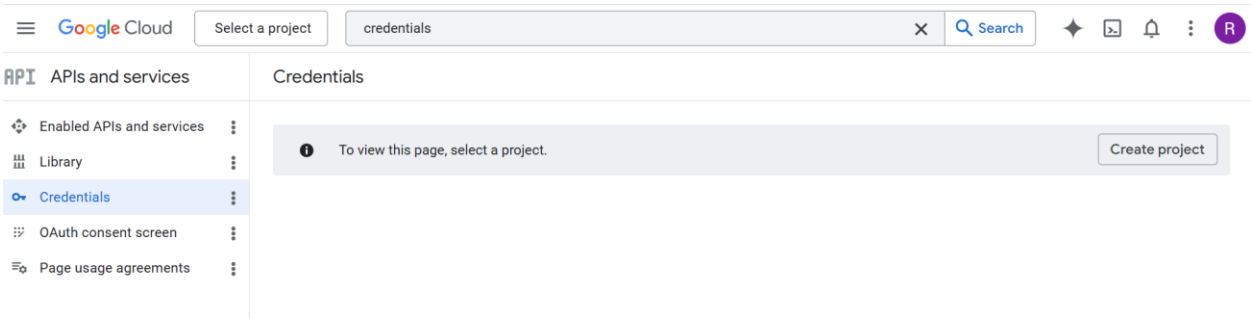
1. Make sure **Apache and MySQL** are running in XAMPP.
2. Place your project in the XAMPP htdocs folder:

C:\xampp\htdocs\oauth-lab\

3. Create the following PHP files in oauth-lab:
    - index.php
    - oauth2callback.php
    - login.php
    - logout.php

Part 2: Register Your Application with Google

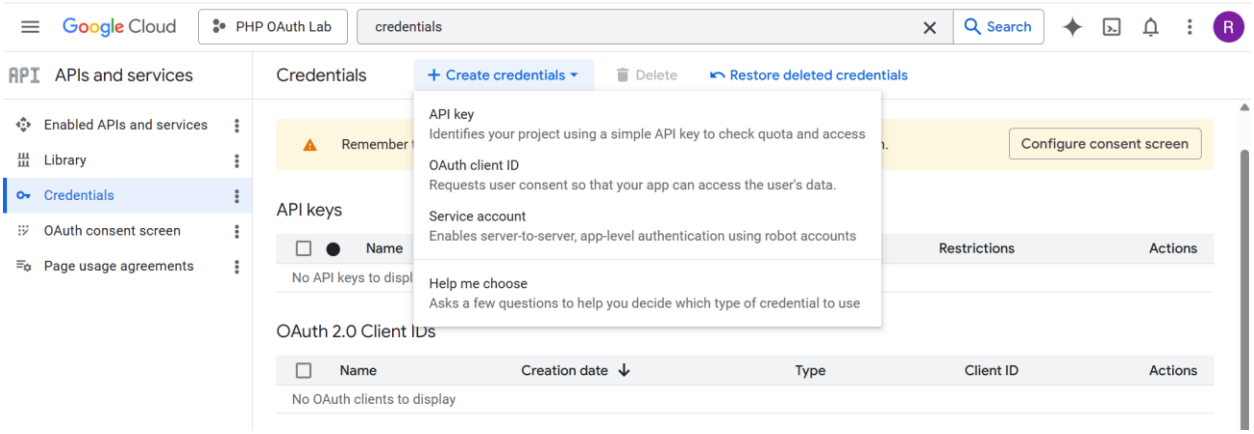
1. Go to Google Cloud Console.
2. Search for API & Services > Credentials



3. Go to Create Project

Project Name: OAuth Lab

4. Go to Create Credentials > OAuth client ID



5. Click > Configure Consent Screen

Application Type: Web Application

Use the same project Name: Oauth Lab

6. Add **Authorized redirect URI**:

http://localhost/oauth-lab/oauth2callback.php

7. Save the credentials and **note your Client ID and Client Secret**.

Part 3: Create PHP Web Pages

1. **index.php**

```php
<?php
session_start();

?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>PHP OAuth Lab</title>
</head>
<body>
  <h1>Login with Google</h1>
  <?php if(isset($_SESSION['user'])): ?>
      <p>Hello, <?php echo $_SESSION['user']['name']; ?>!</p>
      <p>Email: <?php echo $_SESSION['user']['email']; ?></p>
      <a href="logout.php">Logout</a>
  <?php else: ?>
      <a href="login.php">Login with Google</a>
  <?php endif; ?>
</body>
</html>
```

2. **login.php**

```php
<?php
session_start();

$client_id = 'YOUR_CLIENT_ID';
$redirect_uri = 'YOUR REDIRECT URI;
$scope = urlencode('https://www.googleapis.com/auth/userinfo.profile
https://www.googleapis.com/auth/userinfo.email');

$auth_url =
"https://accounts.google.com/o/oauth2/v2/auth?response_type=code&client_id=$client_id&redirect_uri=$redire
ct_uri&scope=$scope&access_type=offline";

header('Location: ' . $auth_url);
exit;
```

3. **logout.php**

```php
<?php
session_start();
session_destroy();
header('Location: index.php');
exit;
```

4. **oauth2callback.php**

```php
<?php
session_start();


// Google OAuth credentials
$client_id ='YOUR_CLIENT_ID';
$client_secret = 'YOUR_CLIENT_secret';
$redirect_uri = 'YOUR REDIRECT URI;

if (isset($_GET['code'])) {
    $code = $_GET['code'];

// Exchange code for access token

    $token_url = 'https://oauth2.googleapis.com/token';
    $post_fields = [
        'code' => $code,
        'client_id' => $client_id,
        'client_secret' => $client_secret,
        'redirect_uri' => $redirect_uri,
        'grant_type' => 'authorization_code'
    ];

    $ch = curl_init();
    curl_setopt($ch, CURLOPT_URL, $token_url);
    curl_setopt($ch, CURLOPT_POST, true);
    curl_setopt($ch, CURLOPT_POSTFIELDS, http_build_query($post_fields));
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);


    // Ignoring SSL for XAMPP/localhost

    curl_setopt($ch, CURLOPT_SSL_VERIFYPEER, false);
    curl_setopt($ch, CURLOPT_SSL_VERIFYHOST, false);

    $response = curl_exec($ch);
    if (curl_errno($ch)) {
        die("Curl error (token request): " . curl_error($ch));
    }
    curl_close($ch);

    $token_data = json_decode($response, true);
```

```
    if (!isset($token_data['access_token'])) {
        die("No access token received:<br><pre>$response</pre>");
    }

    $access_token = $token_data['access_token'];

 //Fetch user info

    $ch = curl_init();
    curl_setopt($ch, CURLOPT_URL,
'https://www.googleapis.com/oauth2/v1/userinfo?alt=json');
    curl_setopt($ch, CURLOPT_HTTPHEADER, ['Authorization: Bearer ' .
$access_token]);
    curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);

    // Ignoring SSL
    curl_setopt($ch, CURLOPT_SSL_VERIFYPEER, false);
    curl_setopt($ch, CURLOPT_SSL_VERIFYHOST, false);

    $user_info = curl_exec($ch);
    if (curl_errno($ch)) {
        die("Curl error (user info request): " . curl_error($ch));
    }
    curl_close($ch);

    $_SESSION['user'] = json_decode($user_info, true);


 // Redirect to index.php after login

    header('Location: index.php');
    exit;

} else {
    echo "Error: No authorization code received.";
}
```

Part 4: Running the Lab

1. Go to your web browser
2. Navigate to your project:

   http://localhost/oauth-lab/
3. Click **Login with Google**.
4. Authorize the app.
5. You should see your **name and email** displayed.
6. Click **Logout** to test session clearing.